

End-to-End MLOps Project Report: Heart Disease Risk Prediction

Executive Summary

End-to-end MLOps solution for heart disease risk classification using the UCI Cleveland Heart Disease dataset. It covers data acquisition, EDA, preprocessing, model development, MLflow, automated tests, CI/CD, FastAPI serving, Docker, Kubernetes, and monitoring.

Dataset Description

The dataset contains patient health variables and a target field. The target is converted to binary form: 0 for no disease and 1 for disease present.

Data Acquisition and Cleaning

The ingestion script downloads the dataset, assigns headers, handles missing values represented by ?, and saves cleaned data.

Exploratory Data Analysis

EDA includes missing values, descriptive statistics, class balance, histograms, and correlation heatmap.

Feature Engineering

Numerical features are imputed and scaled. Categorical features are imputed and one-hot encoded using a reproducible sklearn pipeline.

Model Development

Logistic Regression and Random Forest are trained and evaluated with accuracy, precision, recall, ROC-AUC, and cross-validation.

Experiment Tracking

MLflow logs parameters, metrics, plots, and model artifacts.

API and Containerization

FastAPI exposes /health, /predict, and /metrics. Docker provides reproducible serving.

CI/CD

GitHub Actions runs linting, tests, training, Docker build, and artifact upload.

Deployment and Monitoring

Kubernetes manifests deploy the API. Prometheus and Grafana provide monitoring.

Conclusion

The project is reproducible, automated, containerized, deployable, and monitorable.

Appendix: Key Commands

```
python -m src.data_ingestion
python notebooks/01_eda.py
python -m src.train
pytest tests/ -v
docker build -t heart-disease-api:latest .
docker run -p 8000:8000 heart-disease-api:latest
kubectl apply -f k8s/
```